

Econometrics II

Week 1

Institute of Economic Studies
Faculty of Social Sciences
Charles University in Prague

Fall 2026

Welcome :-)

- This is **Econometrics II** course
- Second in the undergraduate Econometrics sequence
- Taught by **Barbara Pertold-Gebicka**
- Seminar leaders:
 - Veronika Plachá
 - Oliver Škultéty
 - Kai Wang
- The course webpage: [Econometrics II in SIS](#)
 - All course information
 - Materials for lectures & seminars

Outline of Today's talk

- General information about the course
- Syllabus
- Goals of Econometrics II
- Repetition: unbiasedness, consistency, and efficiency

The course

- Econometrics I $\rightarrow$ Econometrics II
- We will use the book
(available in the IES library or online for IES students):

Text book

Jeffrey M. Wooldridge: Introductory Econometrics. A Modern Approach. Boston: Cengage Learning, edition fourth or higher.

- For those keen to study Econometrics in more detail or from another angle, these are the recommended complimentary books
 - "Introductory Econometrics for Finance" by Chris Brooks
 - "Mostly Harmless Econometrics: An Empiricist's Companion" by Angrist and Pischke
 - "Microeconometrics: Methods and Applications" by Cameron and Triveldi

Course organization

■ Lectures

- Wednesdays, 12:30-13:50.
- Students required to **prepare for the lecture**
 - Go through the associated reading
 - Revise the pre-required material
- During lectures I will give examples and explain more difficult material

■ Seminars

- Wednesdays: 15:30-16:50 (room 016); Wednesdays: 18:30-19:50 (room 016); Thursdays: 14:00-15:20 (room 016).
- Each seminar relates to a lecture.
- Students **are required** to attend a lecture or carefully read associated reading before participating in a seminar.
- You will solve sample problems illustrating the topic; problems are similar to exam questions.

Requirements

Course requirements

- 4 Assignments: max. 40 points (10 points each)
- Midterm Exam: max. 20 points
- Final Exam: max. 40 points (at least 21 points to pass the Course)

Grading

- 90.5 - 100 points Grade A
- 80.5 - 90 points Grade B
- 70.5 - 80 points Grade C
- 60.5 - 70 points Grade D
- 50.5 - 60 points Grade E
- 0 - 50 points Grade F (Fail)

Course organization - further remarks

■ Software

- In seminar sessions you will use one software
- R - please install it on your computers!
- In home assignments, you are free to use any software

■ Assignments

- 4 home assignments
- Posted in SIS approximately two weeks before the deadline
- Work in groups of 2 students
- You can earn up to 10 points for each home assignment

Home assignments

- There will be four home assignments during the semester
- Dates
 - First assignment announced Oct 12, deadline Oct 26
 - Second assignment announced Oct 26, deadline Nov 9
 - Third assignment announced Nov 16, deadline Nov 30
 - Fourth assignment announced Nov 30, deadline Dec 14
- Contents
 - Each assignment related to one topic
 - One theoretical problem and one empirical problem
 - The empirical problem should be approached as a small project. Students are expected to write a short report.
- Grading
 - We will give you a detailed feedback within 10 working days.

Exams

- Midterm exam
 - One-shot exam, no additional terms
 - **November 11, 2026, 12:30-3:00pm.**
 - Written during lecture time
 - Multiple choice questions and YES/NO questions
- Final exam
 - must-take exam, three trials possible
 - During the exam period (dates announced soon)
 - **Must obtain more than 50% points in the final to pass the Course!**
 - true/false questions, open theoretical questions, open practical questions
- Grading
 - Score available within 7 working days.

Goals of the Econometrics II course

- **Econometrics I** taught you the basic tools
 - Ordinary Least Squares (OLS) with cross-sectional data
 - Assumptions necessary for OLS estimates to be BLUE
 - Evaluation of regression analysis - R-squared, hypothesis testing
 - Detecting and dealing with heteroskedasticity
 - Model specification and data issues
- **Econometrics II** will extend your knowledge
 - Regression analysis with time series data
 - Regression analysis with panel data
 - What happens and what can/can't we do when OLS assumptions are not satisfied?
 - Regression analysis with limited dependent variables (count variables, binary variables)
 - **Practical application of econometric analysis**

Goals of the Econometrics II course

- After taking **Econometrics II** you should be able to independently conduct an empirical analysis
 - Translate a research question into a regression equation.
 - Chose the best data structure and econometric technique.
 - Conduct regression analysis using a statistical software.
 - Verify whether the obtained estimates can be trusted, adjust your approach if necessary.
 - Interpret your results and/or propose improvements
- The statistics behind econometric methods is an exact science, the data and processes we analyze are not exact.
- One needs to understand the data generating processes in order to correctly apply econometrics.
- We will put a lot of attention to this part of the **art of econometrics**.

How to achieve this goal

- Material covered by Econometrics II is **more difficult** than Econometrics I
- We will learn many new approaches...
- ...which are useful in specific cases, but not in others.
- This requires systematic work during the whole semester
 - Go through the relevant reading **before** each lecture (reading for next week given at the end of each lecture)
 - Revise previously learned skills when necessary (info about revisions given at the end of preceding lecture)
 - During the lecture I will quickly go through the basic/easy stuff
 - We will devote most of the time to practical examples and more difficult material

For the next lecture

- Revise basics of Econometrics I with special attention to
 - Gauss-Markov assumptions for OLS regression on cross-sectional data
 - Properties of OLS estimator on cross-sectional data
 - Unbiasedness and consistency
 - Efficiency
- Read chapter 10 of Wooldrige textbook

Let me help you with some revision

Given the following regression model:

$$y_i = \beta_0 + \beta_1 \cdot x_i + u_i$$

The OLS estimator of the slope parameter (β_1) is:



Let me help you with some revision

Given the following regression model:

$$y_i = \beta_0 + \beta_1 \cdot x_i + u_i$$

The OLS estimator of the slope parameter (β_1) is:

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

Note: All sums go from $i=1$ to $i=n$, i.e. sum over all observations.

Note: A simple (one-explanatory-variable) regression model is given as an example. For multiple regression model it's more convenient to work with matrix notation - why?

Unbiasedness

Theorem: Unbiasedness of OLS (after Wooldridge)

Under Assumptions MLR.1 through MLR.4, the OLS estimators are unbiased estimators of the population parameters: $E(\hat{\beta}_j) = \beta_j, j = 0, 1, \dots, k$.

In other words: an estimator is unbiased when, on average, it provides estimates that are equal to the true parameter it is trying to estimate.

- What does it mean in practice?
- What are the assumptions MLR.1 - MLR.4?

Unbiasedness

What does it mean in practice?

- An unbiased estimator does not systematically overestimate or underestimate the true parameter.
- CAREFUL: unbiasedness does not guarantee that a single estimate is close to the true parameter. Why?

Classical Linear Model Assumptions

Numbering according to Wooldridge:

- Assumption MLR.1 (Linear in Parameters)
- Assumption MLR.2 (Random Sampling)
 - This assumption is very important! When it is satisfied, we have data that can be used to estimate model parameters, the data is representative of the studied population and **observations are independent of each other.**
- Assumption MLR.3 (No Perfect Colinearity)
- Assumption MLR.4 (Zero Conditional Mean)
 - We can also call it the **exogeneity** assumption

Unbiasedness

Where are the MLR assumptions used?

$$\hat{\beta}_{1,OLS} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})} = \dots = \beta_1 + \frac{\sum_{i=1}^n (x_i - \bar{x})(u_i)}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$E[\hat{\beta}_{1,OLS}|X] = \beta_1 + E\left[\frac{\sum_{i=1}^n (x_i - \bar{x})(u_i)}{\sum_{i=1}^n (x_i - \bar{x})^2} | X\right]$$

$$E[\hat{\beta}_{1,OLS}|X] = \beta_1 + \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} E\left[\sum_{i=1}^n (x_i - \bar{x})(u_i) | X\right]$$

$$E[\hat{\beta}_{1,OLS}|X] = \beta_1 + \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \sum_{i=1}^n E[(x_i - \bar{x})(u_i) | X]$$

$$E[\hat{\beta}_{1,OLS}|X] = \beta_1 + \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \sum_{i=1}^n (x_i - \bar{x}) E[u_i | X]$$

$$E[\hat{\beta}_{1,OLS}|X] = \beta_1$$

Detailed video (by Ralph Becker):



Consistency

Theorem: Consistency of OLS (after Wooldridge)

Under Assumptions MLR.1 through MLR.4, the OLS estimator $\hat{\beta}_j$ is consistent for β_j , for all $j = 0, 1, \dots, k$.

Consistency means that if sample size increases indefinitely, the estimator's value converges in probability to the true parameter.

In other words: when sample size increases, the estimator is more likely to provide estimates that are very close to the true population parameter.

Consistency

- An unbiased estimator is consistent as long as its variance tends to zero with increasing sample size
- MLR.4 (after Wooldridge) is actually unnecessarily strong for proving consistency.
- MLR 4' (after Wooldridge, a weaker version) is sufficient:
 $E[u] = 0$ and $Cov(x_j, u) = 0$, for all $j = 0, 1, \dots, k$
- It is possible that an estimator is biased but consistent.

More on asymptotic theory (by Colin Cameron):



Consistency

Where are the MLR assumptions used?

$$\begin{aligned}\hat{\beta}_{1,OLS} &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})} = \dots = \beta_1 + \frac{\sum_{i=1}^n (x_i - \bar{x})(u_i - \bar{u})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \\ &= \beta_1 + \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(u_i - \bar{u})}{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}\end{aligned}$$

$$plim \hat{\beta}_{1,OLS} = \beta_1 + plim \left\{ \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(u_i - \bar{u})}{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2} \right\}$$

$\Rightarrow$ (by the Law of Large Numbers)

$$plim \hat{\beta}_{1,OLS} = \beta_1 + \frac{Cov(x, u)}{Var(x)}$$

Classical Linear Model Assumptions

Numbering according to Wooldridge:

- Assumption MLR.1 (Linear in Parameters)
- Assumption MLR.2 (Random Sampling)
- Assumption MLR.3 (No Perfect Colinearity)
- Assumption MLR.4 (Zero Conditional Mean)
- Assumption MLR.5 (Homoskedasticity)

Gauss-Markov Theorem (after Wooldridge)

Gauss-Markov Theorem

Under MLR.1 through MLR.5, the OLS estimators are the best linear unbiased estimators (BLUE) conditional on X .

Classical Linear Model Assumptions

- Assumption MLR.1 (Linear in Parameters)
- Assumption MLR.2 (Random Sampling)
- Assumption MLR.3 (No Perfect Colinearity)
- Assumption MLR.4 (Zero Conditional Mean)
- Assumption MLR.5 (Homoskedasticity)
- Assumption MLR.6 (Normality)

Note: Never refer to these assumptions by numbers when no context is given!
Rather describe them, refer to their names.

Different textbooks use different numbering. See Greene, page 57 or
Stock & Watson, page 227

Inference Under the Classical Linear Model (CLM) Assumptions

Theorem: Normal Sampling Distribution (after Wooldridge)

Under MLR.1 through MLR.6, the OLS estimators are normally distributed, conditional on X . Further, under the null hypothesis, each t statistic has a t distribution, and each F statistic has an F distribution.

What will we do in Econometrics II

- How can we adjust the OLS estimation method to time-series data?
- How can we adjust the OLS estimation method to panel data?
- What to do when some of the CLM assumptions are not satisfied?
- Nonlinear estimation methods for specific types of dependent variables.

This is all for today!

- Tutorial sessions this week
 - practical revision of Econometrics I
 - how to produce a report based on econometric analysis

Thank you for your attention!

... and do not forget to read Chapter 10 for the next week!